EV Battery Course — Module 4: Evolution of EV Battery Technology

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EV batteries went from about 35 Wh/kg (lead-acid) to about 300 Wh/kg (nickel-rich Li-ion) in roughly a century, and pack prices fell about 93% between 2010 and 2025. This module traces each step, separates what is commercial from what is still in pilot or the lab, and looks at what could realistically change in the next 5–15 years.

1. Learning objectives

- Describe each battery generation used in EVs: chemistry, energy, cost, safety, maturity and its limiting problem.
- Classify any new battery claim as laboratory, pilot or commercial.
- Explain why Li-ion displaced every earlier chemistry.
- Judge which next-generation technologies could affect the EV market in 5–15 years, and how uncertain those forecasts are.

2. Fundamentals: what makes a chemistry win

A battery chemistry wins in EVs when it clears five hurdles at the same time: enough energy per kg for useful range, enough power, enough cycle life, acceptable safety, and low enough cost at scale. History is a sequence of chemistries that cleared some hurdles but failed one.

3. Generation by generation

Generation	Typical cell Wh/kg	Strength	What held it back	Example vehicles
Lead-acid (1859; EVs c.1900, revived 1990s)	30–40	Cheap, robust, recyclable	Very heavy; about 300–500 deep cycles	Early 1900s EVs; GM EV1 Gen 1 (1996)
Nickel-cadmium (1990s EVs)	40–60	Long life, tough	Toxic cadmium; memory effect; cost	Peugeot 106 Électrique, Citroën Saxo Électrique

Generation	Typical cell Wh/kg	Strength	What held it back	Example vehicles
Nickel-metal hydride (1990s–)	60–80	Safe, durable; perfect for hybrids	High self-discharge; heat; 1.2 V/cell	Toyota Prius (1997), RAV4 EV, GM EV1 Gen 2
Early Li-ion: LCO, LMO (2008–2015)	100–200	3× the energy of NiMH	Cost, cobalt, LMO calendar life, thermal safety	Tesla Roadster (2008, LCO 18650s); Nissan Leaf (2010, LMO blend)
Modern Li-ion: NCA, NMC, LFP (2012–today)	160–300	Range of 400–600+ km; costs below $100/kWh at pack level	Ni/Co supply; LFP energy; China supply dominance	Tesla, Hyundai E-GMP, BYD Blade, Tata Nexon EV
Silicon-enhanced anodes (2020s)	250–350	More energy, faster charging	Swelling, cycle life	Silicon-blend premium cells
Sodium-ion (mid-2020s)	120–175	No lithium, nickel or cobalt; cold performance	Lower energy; early supply chain	CATL and BYD sodium cells, early vehicles in China
Solid-state (late 2020s–2030s)	350–500 (targets)	Non-flammable electrolyte; lithium-metal anode	Manufacturing at scale; interfaces; cost	Pilot lines: Toyota, Samsung SDI, QuantumScape
Lithium-sulfur	400–600 (targets)	Cheap sulfur; very high theoretical energy	Polysulfide shuttle; short cycle life	Pilot; drones and aviation first
Lithium-metal (liquid electrolyte)	400+ (targets)	Highest anode capacity	Dendrites; cycle life	Pilot (e.g. SES AI)
Other (lithium-air, magnesium, zinc, multivalent)	Varies	Theoretical potential	Fundamental science unresolved	Laboratory

4. Timeline and maturity

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timeline
  title EV battery technology by maturity, 2026
  Commercial : Lead-acid : NiCd : NiMH : LCO / LMO : NCA / NMC : LFP :
Silicon-graphite blends
  Early commercial : Sodium-ion : LMFP : Semi-solid-state (limited volumes)
  Pilot production : Sulfide all-solid-state : Lithium-metal : Lithium-sulfur
: Silicon-dominant anodes
  Laboratory : Lithium-air : Magnesium and multivalent : Organic electrodes
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How to classify any battery announcement

Stage	What it means	Evidence to look for
Laboratory	Coin cells or small pouches, often a few mAh	Journal paper, university press release, "breakthrough" headline
Pilot	Automotive-format cells made on a pilot line, sampled to OEMs	A-, B- or C-samples; pilot-line capacity in MWh; OEM test partnerships
Commercial	Cells in vehicles sold to customers at volume	GWh-scale factory, model and VIN, warranty

A common trap: an announced factory or a target year is not a commercial product. Many announced dates have slipped.

5. Step-by-step: why each transition happened

1. **Lead-acid to nickel-based:** about twice the energy per kg; same water-based electrolyte limits cell voltage (about 1.2–2 V).
2. **NiMH to Li-ion:** organic electrolyte lets cells reach about 3.6 V and intercalation gives long life. Energy roughly doubled again.
3. **LCO/LMO to NMC/NCA:** less cobalt, more nickel; higher capacity and better life.
4. **NMC to LFP for mass-market:** cell-to-pack design and cheaper materials made LFP's lower energy acceptable; safety and life were better.
5. **Next:** sodium-ion attacks cost and mineral risk; solid-state attacks energy and safety; silicon and lithium-metal attack anode capacity.

6. Engineering equations

Cell mass for a target energy:

$$m_{cells} = \frac{E_{target}\ (\text{Wh})}{e_{specific}\ (\text{Wh/kg})}$$

Compound annual rate of change (for cost or density trends):

$$r = \left(\frac{X_{end}}{X_{start}}\right)^{1/n} - 1$$

n = number of years. A negative r is a yearly decline.

7. Worked examples

Example 1: what a 50 kWh pack weighs in each generation (cells only).

Chemistry	Wh/kg	Cell mass for 50 kWh
Lead-acid	35	1,429 kg
NiMH	70	714 kg
LFP	180	278 kg
NMC	260	192 kg
Solid-state (target)	400	125 kg

A 50 kWh lead-acid battery would weigh more than the rest of a small car.

Example 2: range from a fixed 400 kg cell budget at 150 Wh/km.

- Lead-acid: 400 × 35 = 14 kWh → **93 km**
- NiMH: 400 × 70 = 28 kWh → **187 km**
- LFP: 400 × 180 = 72 kWh → **480 km** (before pack overhead)

This is the core reason EVs became practical only with Li-ion.

Example 3: the rate of cost decline. BNEF puts average pack prices at about $1,474/kWh in 2010 (in real 2025 dollars) and $108/kWh in 2025.

- r = (108 ÷ 1,474)^(1/15) − 1 = 0.0733^0.0667 − 1 = **−16% per year** on average.

8. Real-world data: battery prices today

BloombergNEF's 2025 survey (published 9 December 2025) reported:

Measure (volume-weighted average, 2025)	Price
All Li-ion packs, all segments	$108/kWh (down 8% from 2024)
BEV packs	$99/kWh
LFP packs, all segments	$81/kWh
NMC packs, all segments	$128/kWh
Average in China	$84/kWh
Cells, all segments	$74/kWh

BNEF attributes the fall to cell overcapacity, competition and the shift to LFP, despite higher lithium and cobalt prices in 2025; it expects about $105/kWh in 2026. Sources: BNEF press release, Energy-Storage.news.

9. Outlook: what could matter in 5–15 years

Technology	Likely impact by ~2030	Likely impact by ~2040	Confidence
LFP and LMFP	Dominant in mass-market and 2W/3W	Still large	High
High-nickel NMC + silicon	Premium long-range	Shared with solid-state	High
Sodium-ion	Niche to meaningful in low-cost cars, 2W, storage	Possibly large in low-cost segments	Medium
Semi-solid / all-solid-state	Low-volume premium models	Possibly mainstream in premium	Low to medium
Lithium-sulfur, lithium-metal	Aviation, drones, niche	Uncertain	Low
Lithium-air, multivalent	None	Uncertain	Very low

Why long-range predictions are uncertain: scale-up often takes 10+ years from lab to car; incumbent Li-ion keeps improving and getting cheaper, raising the bar; and raw material prices and policy (export controls, local-content rules) can change the economics quickly.

10. Manufacturing applications

- **Equipment reuse:** sodium-ion and LMFP reuse most Li-ion lines; this speeds their adoption. Sulfide solid-state needs new dry, inert-gas processing and high-pressure stacking, which slows it.
- **Capex risk:** a gigafactory is a 15–20 year asset. Chemistry flexibility in coating, formation and assembly reduces stranded-asset risk.
- **India context:** cell localisation decisions made now (mostly LFP and NMC) must allow for a likely sodium-ion shift in low-cost segments.

11. Common misconceptions

- **"Solid-state will replace Li-ion soon."** Solid-state is a type of lithium battery; even optimistic plans start with low volumes.
- **"A lab result of 500 Wh/kg means 500 Wh/kg cars."** Lab figures often exclude packaging and use tiny cells.
- **"NiMH is obsolete."** It remains in many hybrids because of its durability and safety.
- **"Prices only fall."** Pack prices rose in 2022 when lithium spiked; trends have reversals.

12. Practical case study: choosing a chemistry roadmap for an Indian OEM

Problem. An OEM must commit cell supply for a 2028 compact SUV and a 2030 platform refresh.

Analysis. For 2028, only commercial chemistries are safe: LFP (cost, safety, Indian heat) or NMC (range). For 2030, LMFP and sodium-ion are plausible additions; solid-state is a watch-and-test option.

Decision. LFP cell-to-pack for 2028; design the 2030 pack mechanically and electrically flexible (voltage window, BMS software) to accept LMFP or sodium-ion; run a solid-state sample evaluation without production commitment.

Lesson. Commit to what is commercial; design for options; test what is in pilot.

13. Revision notes

- Each generation roughly doubled energy: lead-acid (35) → NiMH (70) → Li-ion (150–300 Wh/kg).
- Li-ion won through higher voltage (organic electrolyte) and intercalation (long life).

- Pack prices fell about 93% from 2010 to 2025, about 16% a year; 2025 average $108/kWh.
- Classify claims as lab, pilot or commercial; announcements are not products.

14. Glossary

Term	Meaning
Memory effect	Apparent capacity loss in NiCd after repeated partial cycles
Semi-solid-state	Cell with reduced liquid electrolyte plus solid components
A/B/C-sample	Successive automotive prototype maturity stages
Polysulfide shuttle	Li-S degradation where dissolved sulfur species migrate
Dendrite	Needle-like lithium growth that can short a cell

15. Knowledge check

Conceptual

1. Why could water-based chemistries never exceed about 2 V per cell?
2. Why did NiMH survive in hybrids after Li-ion appeared?
3. What evidence distinguishes a pilot-stage battery from a commercial one?
4. Why did LFP return to cars after NMC seemed to have won?
5. Why do lab energy densities overstate vehicle-level values?

Numerical

1. A scooter needs 3 kWh. Compare cell mass using lead-acid (35 Wh/kg) and LFP (170 Wh/kg).
2. Pack prices fell from $139/kWh (2023) to $108/kWh (2025). What is the annual rate?
3. With a 300 kg cell budget and 140 Wh/km, what range do LFP (180 Wh/kg) and a 400 Wh/kg solid-state cell give?

Application

1. A start-up claims a 600 Wh/kg battery "ready for EVs by next year." List five questions you would ask.

2. For India's two-wheeler market, rank LFP, NMC and sodium-ion for 2030 and justify.

16. Answer key

Conceptual

1. Water electrolyses at about 1.23 V; above that it decomposes. Kinetics allow lead-acid to reach about 2 V, but not 3–4 V.
2. Hybrid batteries do many shallow cycles at high power; NiMH is durable, safe, tolerant of abuse and already validated, so it remained cost-effective.
3. Automotive-format cells at volume, a GWh factory, cells in customer vehicles with warranty, versus samples to OEMs from a pilot line.
4. Cell-to-pack integration recovered much of its pack-level energy gap, while its cost, safety and life advantages, plus nickel and cobalt price spikes, made it attractive.
5. Tiny cells, thin electrodes, excess lithium, no casing or pack overhead, and benign test conditions.

Numerical

1. Lead-acid: 3,000 ÷ 35 = **85.7 kg**. LFP: 3,000 ÷ 170 = **17.6 kg**.
2. (108 ÷ 139)^(1/2) − 1 = 0.777^0.5 − 1 = 0.881 − 1 = **−11.9% per year**.
3. LFP: 300 × 180 = 54 kWh → 54,000 ÷ 140 = **386 km**. Solid-state: 300 × 400 = 120 kWh → **857 km** (cell level, before pack overhead).

Application

1. Cell format and capacity? Measured at what C-rate, temperature and depth of discharge? Cycle life to 80% capacity? Independent third-party test data? Pilot-line capacity and cost per kWh? Safety test results (nail, overcharge, thermal propagation)?
2. LFP first (cost, life, heat tolerance, mature Indian supply); sodium-ion second and rising (cost, no lithium, mineral security; energy adequate for city scooters); NMC third (energy not needed, cost and heat disadvantages).

17. Further learning

- IEA *Global EV Outlook* and *Batteries and Secure Energy Transitions* (2024).
- BNEF annual Lithium-Ion Battery Price Survey (press summaries are free).
- J. Janek & W. Zeier, "Challenges in speeding up solid-state battery development," *Nature Energy* 2023.
- Your own deck *EV Battery Roadmap and Strategy*, slides on Gen 1–5, for comparison.

Sources

- BNEF: Lithium-ion battery pack prices fall to $108/kWh (9 Dec 2025)
- Energy-Storage.news on the BNEF 2025 survey